

Com2uS: Connecting with global players with cloud scalability

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ABOUT COM2US

Com2uS, the company behind the mobile game series, Summoners War, has recently migrated its first game in the series, "Summoners War Arena", to Google Cloud in order to respond to the sudden influx of players as the game gets updated with new features and content. After migrating to



About Com2uS

Com2uS is a mobile game company that started in Korea as the first mobile game service in 1999. Since the development of smartphones in the late 2000s, Com2uS has expanded its business overseas with series such as Summoners War, Com2uS Pro Baseball, MLB 9 Innings, and NBA NOW, as it delivers gaming experiences to players from over a dozen countries.

Industries: Games

Google Cloud, the Location: Korea
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Google Cloud (<https://cloud.google.com/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

Cloud Nat (<https://cloud.google.com/nat>)

Compute Engine
(<https://cloud.google.com/compute/>)

Google Cloud results

- Ensure stability in the event of an influx of players through cloud scalability
- Optimize infrastructure management through provisioning servers with internal IP addresses
- Ensure that players globally can connect to its

Migrated over
600 servers to
the cloud while
optimising
network design
and security

games
without
experiencing
connection
delays or
failures

Com2uS (<https://com2us.com/>) has been making mobile games since the days of feature phones. In particular, it has developed a popular mobile role-playing game named Summoners War. The series has been rapidly gaining popularity among gamers, with the first game, Summoners War: Sky Arena, launching in Korea and around the world since 2014. With more than 200 million users, new content is constantly being added to keep the community engaged, with the series establishing itself as a beloved franchise. It has also released its second game, Summoners War: Lost Centuria, and Summoners War: Chronicle.

Recently, Com2uS has decided to migrate the Summoners War series to Google Cloud (<https://cloud.google.com/>). The series was originally hosted in an on-premise data center, with new titles being developed on Google Cloud, beginning from the very first stages of planning. This shift is done to anticipate potential peaks in traffic. For many game developers, public clouds are considered during development as new games can swell in popularity rapidly, especially in the early stages when marketing efforts are most intense. This means that traffic can fluctuate in unpredictable ways. As such,

public clouds like Google Cloud offer the flexibility to accommodate such spikes in traffic.

That said, it is not uncommon for established games like Summoners War, which was released nine years ago, to migrate to the public cloud. This is because Com2uS operates several mobile game services, with new games still being developed.

Com2uS is constantly thinking about how best to redistribute its resources according to the game and market demands. "Summoners War: Sky Arena is constantly being developed over the years, with various features that are added gradually. Even if you use your own cloud, your infrastructure is physically scattered across multiple data centers. Real-time feedback is important for games, and network traffic within the data center as well as external traffic has steadily increased. That's why we started thinking about bringing them under one data center or using a public cloud," says Sang-min Lim, Infrastructure Technology department head at Com2uS.

Lim explains that the continuous growth of mobile games requires adequate infrastructure, stability, flexibility, and scalability to support them. In particular, the number of visitors has increased significantly with frequent content updates, new service launch events, and the SWC Summoners War World Arena Championship, which sees fans from all over the world attending the event. When such events are held, the infrastructure needs to be scaled accordingly to respond to network demands in real-time, but it is not easy to meet this flexibly

with the company's existing hybrid cloud environment.

On Google Cloud, any additional infrastructure resources can be allocated in real time, which is more suitable for its current operations than creating a new data center and its own cloud. While the lift-and-shift approach of moving previously operated servers to Compute Engine (<https://cloud.google.com/compute>) was used, Com2uS has also found that some infrastructure improvements were also required.

"Summoners War: Sky Arena has continued to develop while operating for nine years in the global game market. We have been operating on-premise and in our own cloud as a hybrid, but as the number of servers increased, the responsibilities on management and internal network traffic have gradually increased," says Lim. "We believe that Google Cloud can seamlessly tie them together, which will allow Com2uS to benefit from flexible resource allocation, but also a systematic and consistent infrastructure for development."

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Meeting connectivity requirements with a new network blueprint

Depending on the game's features, country of release and environment, detailed infrastructure design and differentiation are important. These are considerations Com2uS had when deploying its public cloud. Most importantly, ensuring stability in its operations is vital across all the countries that its games are being launched in.

Lim believes that infrastructure flexibility and smooth network performance are the strengths of Google Cloud. Summoners War: Sky Arena has a global reach, and it generates constant traffic in six major markets. Depending on the situation, the traffic may increase by two to three times. In addition, there is a change in network management, which can also lead to connection delays. This is a concern that the company's operational engineers have. Fortunately, Cloud Load Balancing (<https://cloud.google.com/load-balancing>) could

redistribute the rapidly changing traffic without issues, so gamers connecting from anywhere in the world did not have to worry about connection failures between the client and web application servers.

As the service and the number of servers increase, IP management is another challenge. Since Summoners War: Sky Arena is a large-scale service with more than 600 servers, latency and traffic are common issues. Then there is the need to connect servers with individual IPs, from IP resource allocation to system policy, which has made management extremely complex.

"Cloud NAT (<https://cloud.google.com/nat>) by Google Cloud has dramatically reduced our network design concerns," says Lim. "Cloud NAT can freely allocate the IPs of virtual machines as internal IPs, so from the perspective of development or operation, there is only one server visible to the outside, but numerous virtual machines inside it are managed with their own internal IPs."

For data center operations specifically, external IPs are allocated so that each server can be accessed from the outside. However, IPs are not only limited, but also difficult to manage because these are resources that have to be distributed within the company. With Cloud NAT, the server that is hosting Summoners War: Sky Arena and individual servers can be assigned virtual IPs, so IPs can be allocated freely. In addition, address rules can be directly defined according to internal operating policies, so the network becomes more structured than before,

with the role of a server easily identified by looking at its IP address.

Even when Cloud NAT interlocks data between WAS, security gaps are a thing of the past as policies can be set up and operated, so that only designated virtual servers can communicate. From a security perspective, management has become immensely simpler.

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Enhancing the global gaming experience

Lim explains that the most important role cloud can play in a mobile game environment, where the market has expanded globally, is to nurture the best game experience for users from anywhere in the world. Since online games require real-time feedback and stability, the infrastructure cannot compromise on performance and availability.

Com2uS is already developing and deploying other games in the Summoners War series, such as Summoners War: Lost Centuria and Summoners War: Chronicle on Google Cloud. The platform allowed users around the world to play these games without encountering many issues. While seamlessly connecting users across the world, Google Cloud was also able to dynamically scale the infrastructure, even at events with many gamers,

such that there was no unstable connection or poor latency.

The infrastructure, too, can be configured with ease thanks to [Google Cloud Console](https://cloud.google.com/cloud-console)

(<https://cloud.google.com/cloud-console>), which made deploying services and scaling infrastructure a breeze. As resources could be handled in a web-based interface, Com2uS could manage the infrastructure without a separate development environment. At the same time, the company could ensure a high level of security, since Google Cloud is accessed based on Google accounts, which simplified access and permission management.

The unified point of management delivered by Google Cloud console has resulted in improved consistency across the business. The flow of data became clear throughout the entire development process, from source development to distribution and log analysis, with operational policies becoming consistent. Furthermore, productivity has improved due to the unified business management.

Serving as a one-stop solution for mobile game development

By July 2022, Summoners War: Sky Arena was fully migrated to Google Cloud. Since many players are accessing the game from abroad, Com2uS can benefit from the global network of Google Cloud. Even though Com2uS is running Summoners War in the Seoul region of Google Cloud, the global infrastructure of Google Cloud delivers low latency

connections that allow players around the world to experience no delay in their gaming experience.

"Compute Engine has eased development by responding immediately to the rapid increase in users. Recently, while cross-promotional marketing was conducted on the three Summoners War titles, the traffic for each game increased significantly. Compute Engine was able to allocate the necessary resources flexibly in real time," says Lim.

With the rise of mobile games, games in general are becoming commonplace across the world, as they increasingly expand their online capabilities regardless of the way games are delivered in a specific region. In a market where peaks in traffic are difficult to predict, Lim believes that Google Cloud can provide Com2uS with the computing resources they need in any situation, while allowing gamers to quickly connect to the game from anywhere. Moreover, Com2uS also has the capacity to continuously develop new and original content for Summoners War without worrying about infrastructure and operations.

With Summoners War: Sky Arena, Com2uS has demonstrated that the platform is not only suitable for new games, but also for established titles with a constantly evolving virtual world. "At the recent ninth anniversary event of Summoners War: Sky Arena, fans gathered two to three times more than usual, but everyone was able to create a festive atmosphere," says Lim. "Looking back over the past nine years, our game operating infrastructure has steadily evolved from on-premise to our own cloud,

and then to Google Cloud. It is encouraging to see how the way we develop games and our operations evolve along with the game meeting the expectations of global gamers."

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